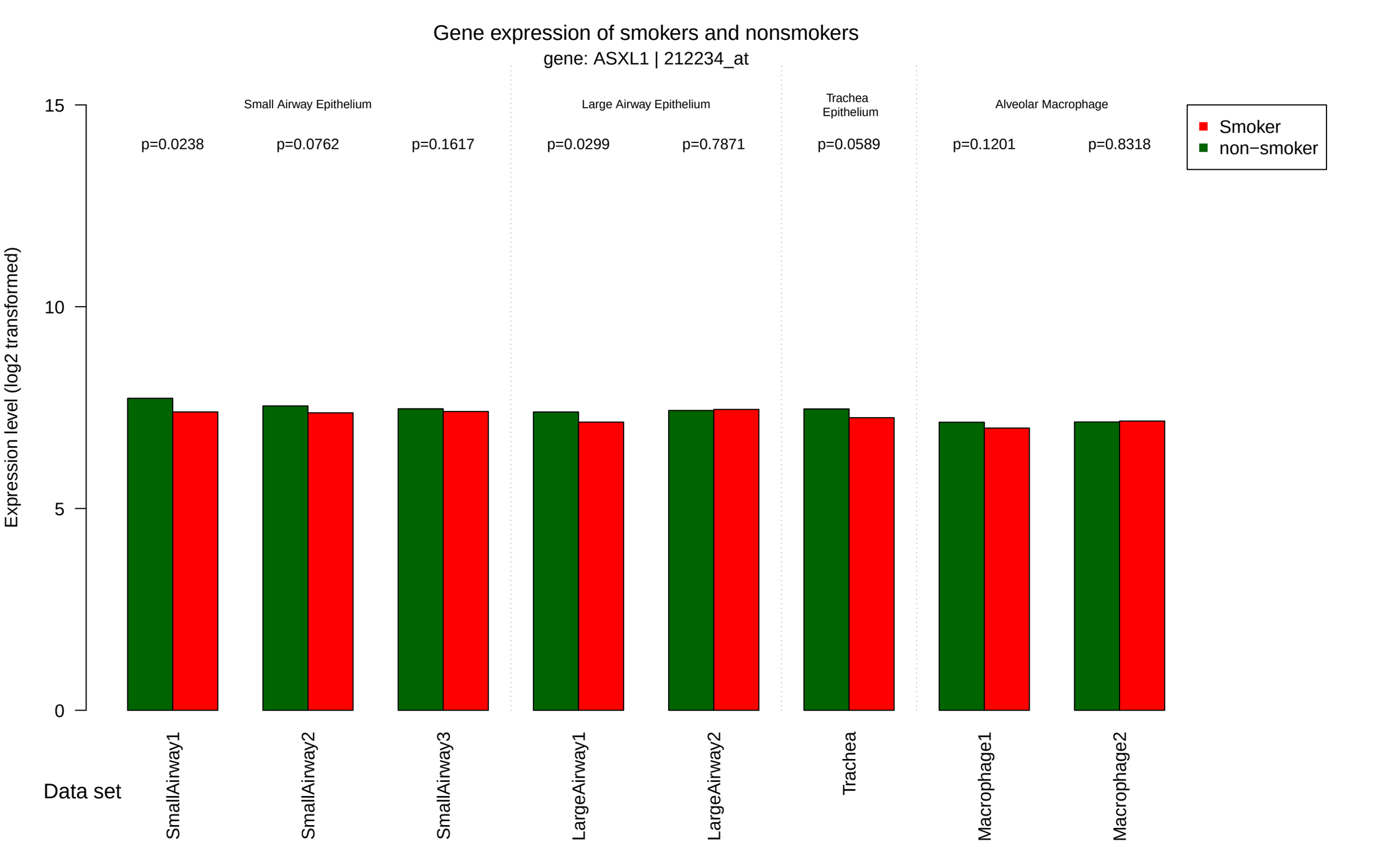
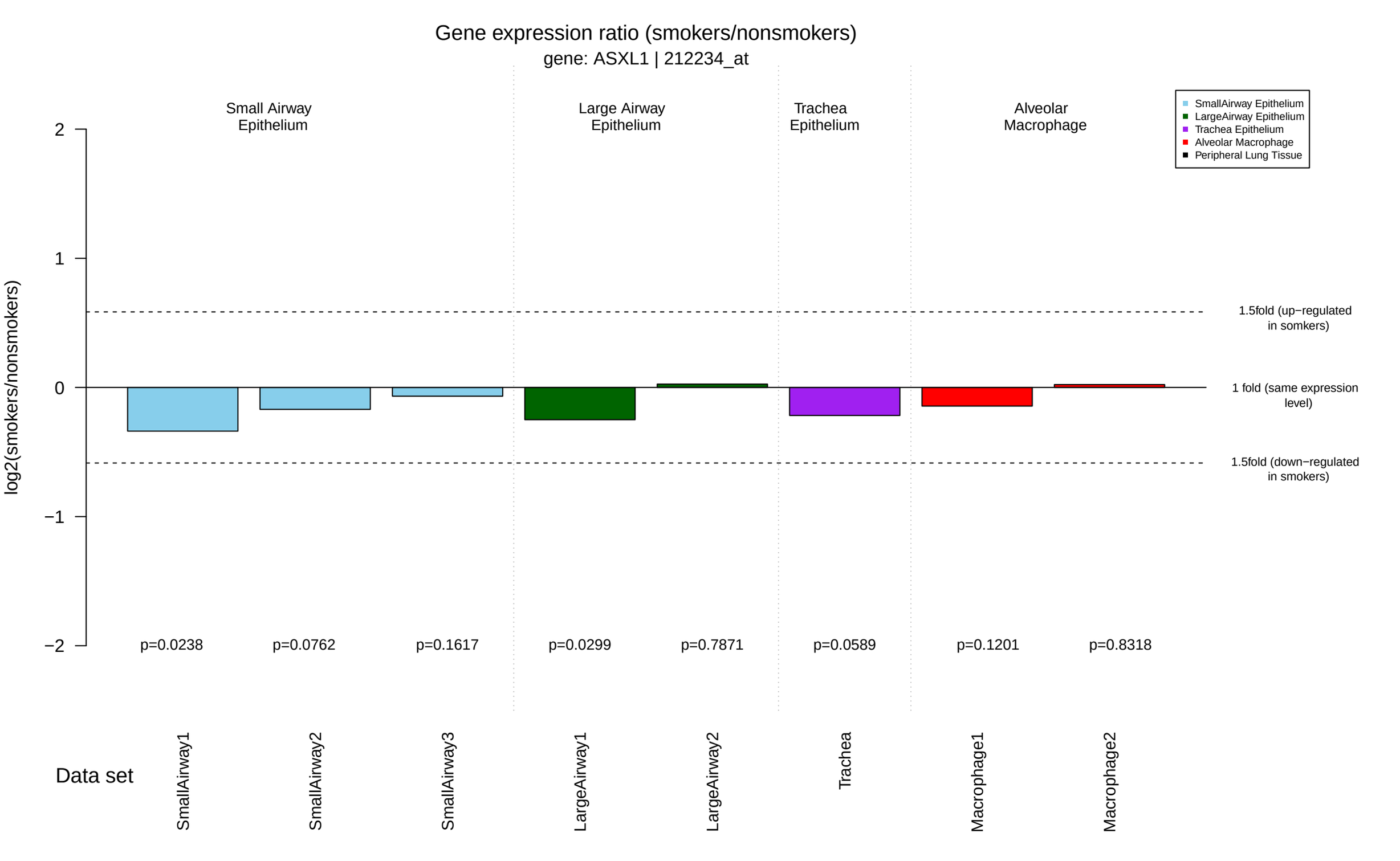
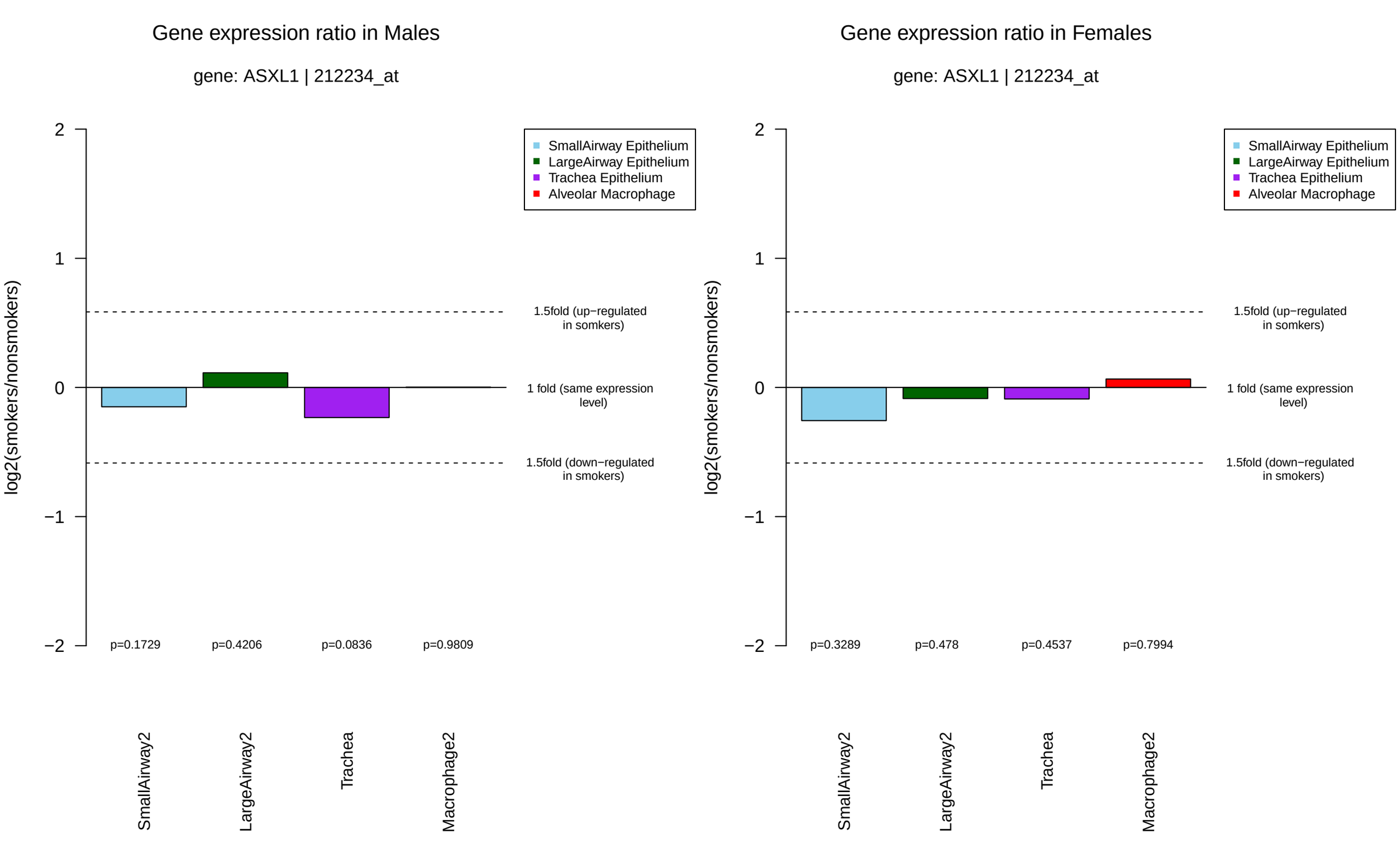
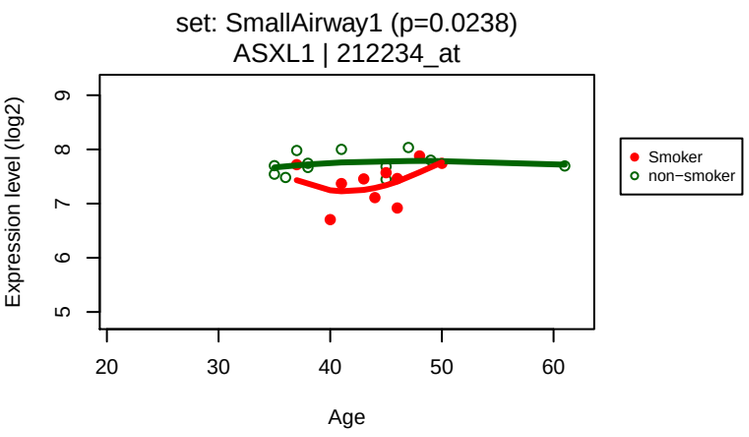
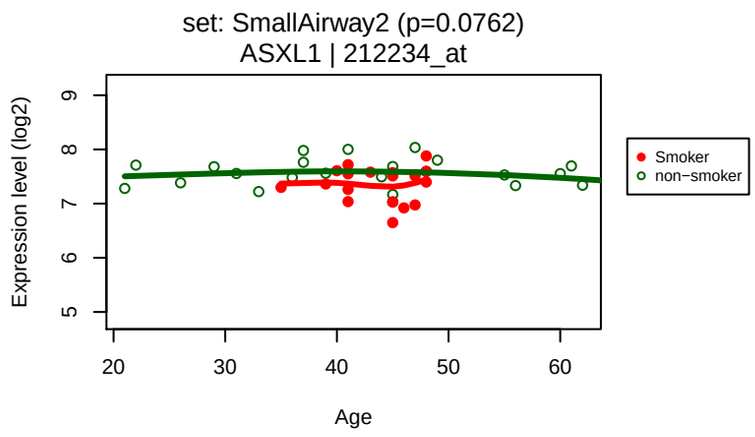
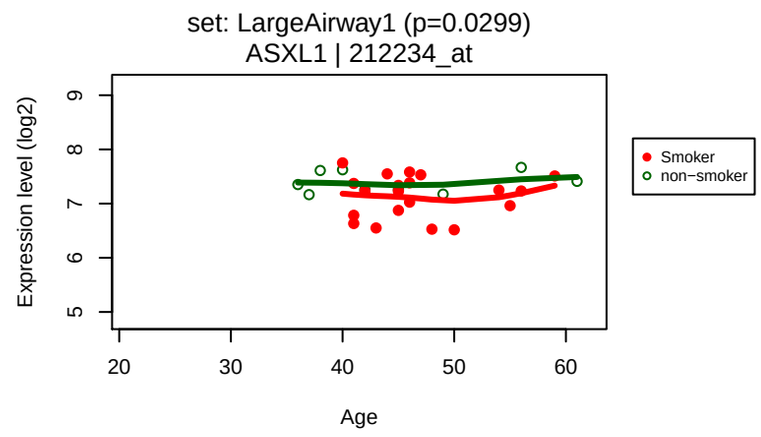

Gene expression of smokers and nonsmokers
gene: ASXL1 | 212234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0238
p=0.0762
p=0.1617
p=0.0299
p=0.7871
p=0.0589
p=0.1201
p=0.8318
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

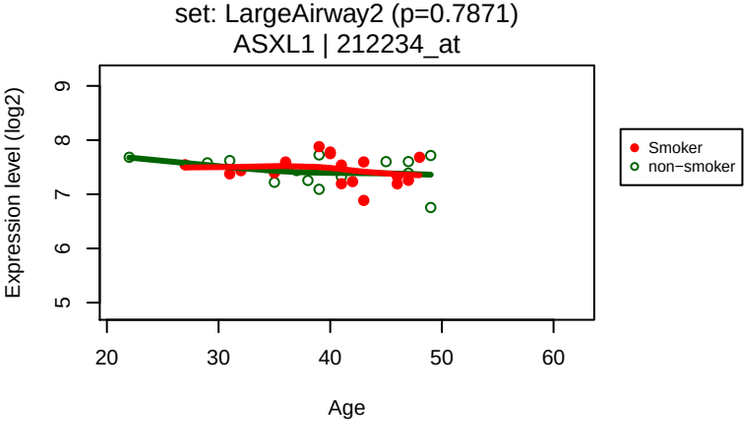

Gene expression ratio (smokers/nonsmokers)
gene: ASXL1 | 212234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0238
p=0.0762
p=0.1617
p=0.0299
p=0.7871
p=0.0589
p=0.1201
p=0.8318
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

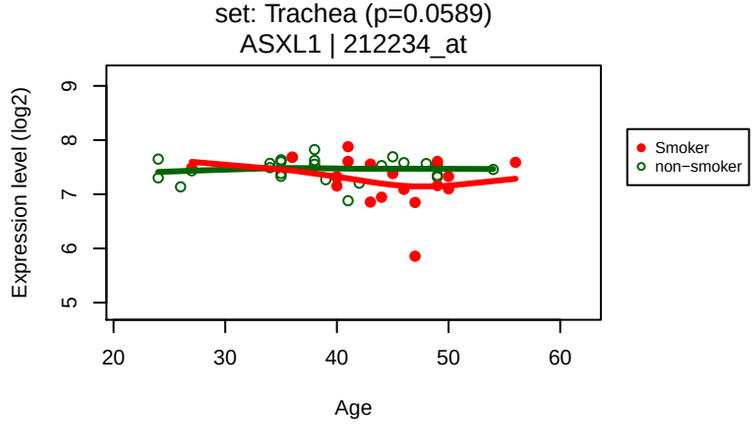

Gene expression ratio in Males
gene: ASXL1 | 212234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1729
p=0.4206
p=0.0836
p=0.9809
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASXL1 | 212234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3289
p=0.478
p=0.4537
p=0.7994
SmallAirway2
LargeAirway2
Trachea
Macrophage2

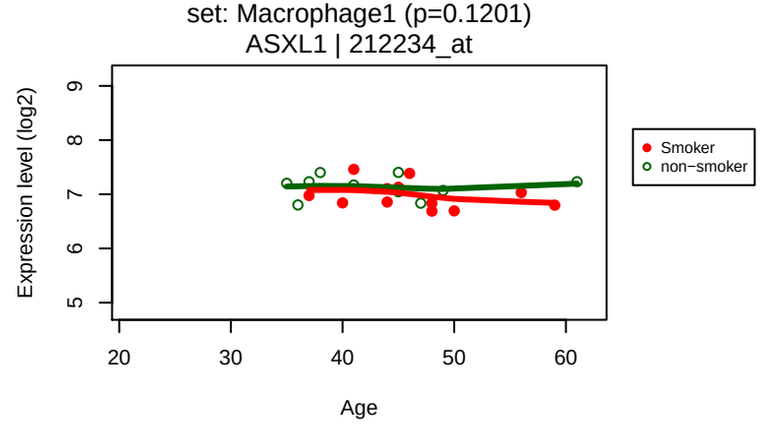

set: SmallAirway1 (p=0.0238)
ASXL1 | 212234_at
Expression level (log2)
Age
Smoker
non−smoker

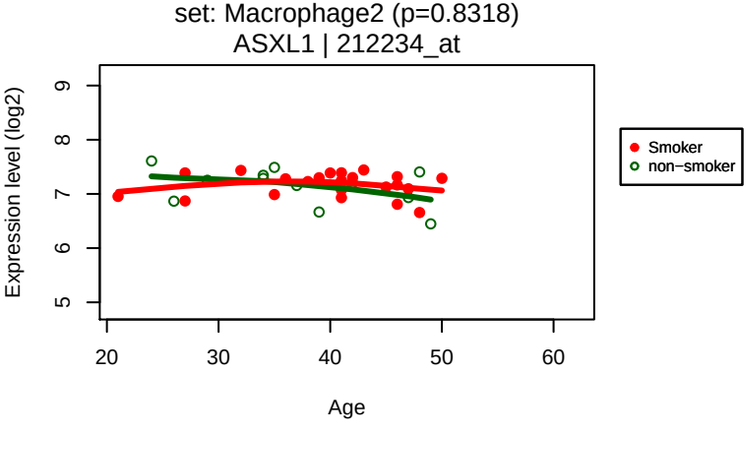

set: SmallAirway2 (p=0.0762)
ASXL1 | 212234_at
Expression level (log2)
Age
Smoker
non−smoker

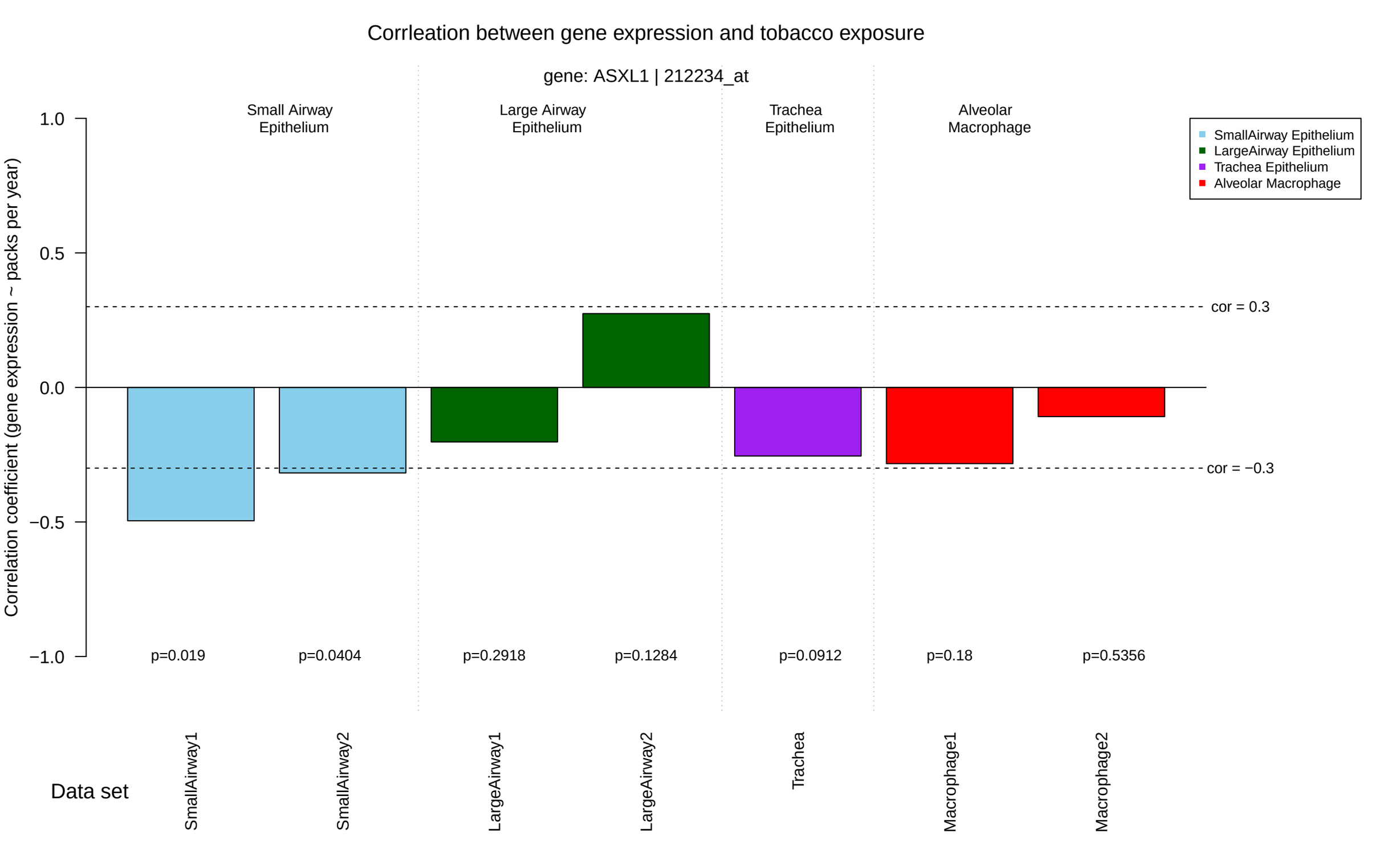

set: LargeAirway1 (p=0.0299)
ASXL1 | 212234_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7871)
ASXL1 | 212234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0589)
ASXL1 | 212234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1201)
ASXL1 | 212234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8318)
ASXL1 | 212234_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASXL1 | 212234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.019
p=0.0404
p=0.2918
p=0.1284
p=0.0912
p=0.18
p=0.5356
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2